
Romanization Flattens the Field: Script and Metric Sensitivity in Sinhala SLM Evaluation

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Abstract

Language model benchmarks for Sinhala are written in the formal script, while the people who would use a model mostly type an ad hoc Romanized register in Latin letters. We study that mismatch for Sinhala, an Indo-Aryan language with over 17 million speakers. A recent benchmark reported models about 312 times worse on Romanized than Sinhala-script text, and no effect of scale. Re-measuring 31 checkpoints, we reproduce those perplexities and find almost all of the ratio is the unit of account: word counts are identical within a parallel pair, and in matched units the median checkpoint assigns only 2.3% more loss to the same content, while 12 of 24 assign less. Measured tokenizer-independently, scale does predict Sinhala-script ability. In the first downstream evaluation of Sinhala script variation, 9,479 items and eleven instruction-tuned checkpoints over 208,538 generations with automatic romanization, the effect is not a uniform drop but a flattening: a 22.4 point spread in Sinhala-script accuracy becomes 6.7, and Romanized accuracy tracks it at a slope of 0.45 across benchmark cells and 0.22 across checkpoints. On offensive-language detection, accuracy barely moves while Matthews correlation loses over half its value. We close with five practices for evaluation under script variation.

1 Introduction

Sinhala is an Indo-Aryan language written in a script of its own and spoken by more than 17 million people, almost all of them in Sri Lanka [1, 2]. Like several South Asian languages it has an everyday orthographic split where formal writing uses the Sinhala script, while comments, chat and social media are dominated by Romanized Sinhala, known locally as Singlish and typed with Latin letters. It has no standard spelling, so කොහොමද (“how are you”) is typed *kohomada*, *kohomd* or *kohomade* depending on the writer.

Every published Sinhala evaluation set we know of is in the Sinhala script. That is an easily missed coverage gap, because the benchmark is in the right language and the wrong register. Romanized registers persist for durable infrastructural reasons rather than as a fashion [3], and in a deployed maternal-health service Khullar et al. [4] found 83% of Hindi and 73% of Telugu user messages Romanized, with models handling them measurably worse. Anyone serving Sinhala users is serving mostly Romanized input while validating on Sinhala-script data.

Rajapakse and Weerasinghe [5] measured that gap first, benchmarking 24 open-weight models with perplexity and reporting a median 312-fold degradation from Sinhala-script to Romanized text and no correlation between parameter count and script handling. Both numbers depend on the metric rather than on the models, because perplexity is normalized by token count and token count for a

fixed string is entirely a property of the tokenizer [6, 7]. Sinhala is exactly where that matters, since tokenizers segment it very unevenly [8, 9].

Recomputing that benchmark in units the two conditions share leaves very little of the reported ratio standing: a median 2.3% rise in loss on identical content, with half the pool cheaper on Romanized text, and the claim that scale does not help does not survive (§3). In the first downstream evaluation of Sinhala script variation across 9,479 items, 208,538 generations with automatically Romanized inputs, and eleven instruction-tuned small language models (SLMs, 1.1B to 14.7B parameters), romanization compresses the range of model quality far more than it shifts the mean, turning a 22.4 point spread in accuracy into 6.7 points (§4). And on offensive-language detection, accuracy falls a few points while Matthews correlation loses more than half its value, so the metric decides whether the damage is visible at all. None of this needed new models or new annotation, only a different denominator and a competence screen, which §5 states as practices.

2 Setup

Data and conditions. Every item is evaluated twice, in the Sinhala script as published and in a Romanized form with identical content, so every comparison is paired and differs only in orthography. The intrinsic corpus is the 500 parallel sentence pairs of Rajapakse and Weerasinghe [5], whose Romanized side native speakers typed, so the intrinsic results do not depend on our transliteration. Downstream we use every item of three sets with no sampling. 6,879 multiple-choice items from SinhalaMMLU [10], where a mix of four and five option items puts guessing at 23.4%, the complete 2,500-item test split of the SOLD offensive-language task [2], 40.6% offensive so the majority class scores 59.4%, and the 100 Sinhala items of Global PIQA [11, 12]. That is 9,479 items over eleven instruction-tuned SLM checkpoints, or 208,538 zero-shot greedy generations in 80 GPU hours, with one prompt template for every dataset so task and prompt are not confounded. The intrinsic sweep covers 31 checkpoints from 350M to 14.7B, 24 of them prior work’s pool (Appendix B).

The Romanized side. Almost every Sinhala transliteration tool runs Latin to Sinhala [13] and we need the reverse, so we wrote a deterministic grapheme-level transliterator using the digraph conventions speakers actually type, turning වෙසක් උත්සවයේදී බෞද්ධයින් into *wesak uthsawayeedii bauddhayin*. Against 755,000 reference items, 455,000 of them human-supplied and the rest machine-augmented, it has the lowest character error rate of four methods, and 88.6% of its spellings are ones a human supplied for that word, against 84.2% for the speakers who typed the same sentences (Appendix H). Its mapping tables predate that corpus and were never revised against it, so the comparison is out of sample. It is released with the evaluation harness and all analysis code at <https://github.com/sehan-dissanayake/sinhala-script-robustness>.

Statistics and the competence screen. Paired comparisons use McNemar’s test [14] with Holm correction over checkpoints within a task [15], and intervals come from 10,000 paired bootstrap resamples. A model already at chance in the Sinhala-script condition cannot lose anything by being given Romanized input, so averaging it in would understate the effect. We mark a checkpoint *competent* on a task when its Sinhala-script score beats the guessing baseline at $p < 0.01$ under a Poisson-binomial null and it does not put more than 90% of its answers on one option. Six of eleven pass on SinhalaMMLU, five on SOLD, and none on Global PIQA. Neither threshold is load-bearing: the SinhalaMMLU six are unchanged under three of four alternative settings (Appendix E).

3 Perplexity was measuring the tokenizer

Bits per byte divides total negative log-likelihood by a unit of the text rather than by its tokenization, $BPB = \sum_s NLL(s) / (\ln 2 \sum_s |s|_{\text{bytes}})$, and likewise per whitespace word, so the denominator does not depend on the vocabulary. Our recomputed perplexities reproduce the published values to a median of 0.05%, so what follows is the metric, not a different run (Appendix C).

The two metrics disagree. Across the 31 checkpoints, Sinhala-script perplexity and bits per byte are statistically unrelated ($\rho = 0.08$, $p = 0.68$), and Gemma-2-9B ranks 26th of 31 on the first and 2nd on the second. Perplexity tracks tokenizer fertility instead ($\rho = -0.87$, $p < 10^{-9}$), which runs from 4.3 tokens per Sinhala word for the Qwen3.5 and Gemma tokenizers to 14.2 for models that fall back

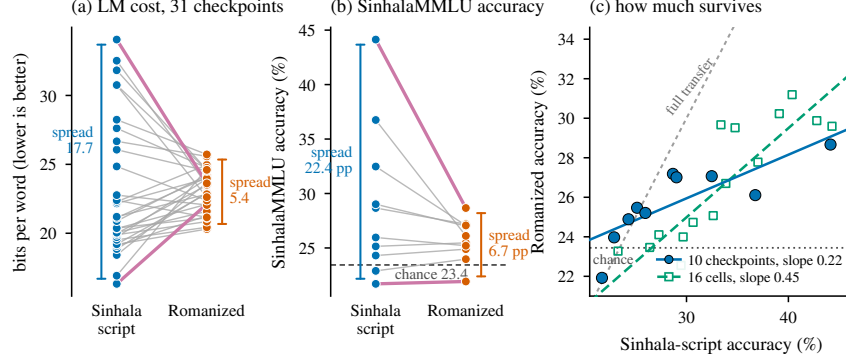


Figure 1: (a) Language modelling cost; word counts are identical within a pair, so the columns are comparable, and highlighted lines are checkpoints cheaper on Romanized text. (b) SinhalaMMLU accuracy, a 22.4 point spread becoming 6.7. (c) The same numbers as levels, with the 16 domain by difficulty cells; both slopes fall well below the full-transfer slope of 1. LaMini-GPT-1.5B is omitted from (b) and (c) for unparseable output on 95.9% of Sinhala-script items.

to bytes (Table 2), and spreading the same loss over three times as many tokens lowers perplexity without the model knowing more.

That makes the reported ratio decomposable rather than arguable. Perplexity is $\exp(\text{NLL}/T)$ for T scored tokens and $\text{NLL} = \ln 2 \cdot A \cdot U$ for A bits per unit over U units of text, so $\log_2 \text{PPL} = Ax$ with $x = U/T$ the units each token carries, and the change across script conditions is exactly

$$\Delta \log_2 \text{PPL} = \underbrace{\bar{A}(x_r - x_u)}_{\text{token-count normalizer}} + \underbrace{\bar{x}(A_r - A_u)}_{\text{per-unit loss}}, \quad (1)$$

with $\bar{A}$ and $\bar{x}$ the two-condition means: the first term is identical content cut into a different number of tokens, the second the model assigning less probability per unit. It is an identity, holds for any unit, and holds to 5×10^{-15} bits here.

On prior work’s 24 checkpoints [5] the median gap is 8.29 bits, which is their 312-fold ratio. Applied per checkpoint and aggregated as a share, the normalizer takes a median 48% of that gap on those 24 and 39% over all 31. The split is not unique, though: the unit has to be chosen, and bytes are not matched across a pair, since Sinhala UTF-8 spends 2.65 bytes per character against 1.00 for Latin and its side of the corpus occupies 2.30 times the bytes. Over whitespace words, whose counts are identical in every pair, the same identity puts 99% of the gap on the normalizer.

So the statement worth quoting needs no unit at all. Because word counts match exactly, the ratio of bits per word is the ratio of total loss over identical content, and on the published 24 its median is 1.02: the median checkpoint assigns 2.3% more loss to the Romanized rendering of the same sentences, and 12 of the 24 assign *less*. Reported and actual degradation are close to unrelated. SmoLM3-3B has the largest ratio of all 31 at 769-fold while assigning 4% less loss to the Romanized text; Gemma-2-9B, genuinely the worst at 46% more loss, reports a middling 357-fold. Almost all of the headline is the unit of account, not half of it.

None of this says the models are fine on Romanized Sinhala; §4 shows they are not. It says a per-token ratio cannot measure it, and that the signal to carry forward is not the size of the gap but the collapse of its range. The scale claim reverses too. Parameter count is uncorrelated with Sinhala-script perplexity ($\rho = 0.15$, $p = 0.42$) but correlated with Sinhala-script bits per byte ($\rho = -0.55$, $p = 0.001$), including on prior work’s own 24 checkpoints ($\rho = -0.42$, $p = 0.04$). What scale does not buy is Romanized ability ($\rho = -0.28$, $p = 0.13$). Family, tokenizer and corpus vary together here, so that is an association and not a scaling result.

4 Model performance under Romanized input

Figure 1(a) shows the effect that organizes the rest. Sinhala-script cost spans 16.3 to 34.1 bits per word across the 31 checkpoints while Romanized cost collapses into a 20.3 to 25.7 band, and a full

bit per word of Sinhala-script improvement buys only 0.14 bits on Romanized text (95% CI 0.05 to 0.25). Which checkpoints end up cheaper on Romanized text is predictable, because the better a checkpoint is on the Sinhala script the more it loses ($\rho = -0.88$, $p < 10^{-9}$). No checkpoint beats a held-out character bigram on Romanized Sinhala, 3.06 bits per byte against 3.29 for the best of the 31 and 3.81 for the median (Appendix D).

The same shape appears downstream. Four of the six checkpoints competent on SinhalaMMLU lose significantly after correction (Appendix Table 7), Qwen3.5-9B falling from 44.1% to 28.7%, a 15.5 point drop (95% CI 14.0 to 16.9), and the two that lose least had 2.5 and 5.2 points of headroom. Pooled over those six, the gap is 6.0 points (95% CI 5.4 to 6.5, $n = 41,274$). In Figure 1(b) no checkpoint is above 28.7% against a 23.4% chance baseline, so the best keeps 5.2 of its 20.7 points of headroom. Regressing Romanized on Sinhala-script accuracy over the benchmark’s 16 domain by difficulty cells, where the units are items, gives a slope of 0.45 (95% CI 0.36 to 0.63, and 0.48 weighted by item count). We take that as the primary estimate: four times the units of the checkpoint-level fit, and none of its confound, since checkpoints differ in family, tokenizer, corpus and tuning at once. Across the ten checkpoints producing parseable output the slope is 0.22 (95% bootstrap CI 0.11 to 0.54, leave-one-out 0.17 to 0.26), falling to 0.12 on the competent six and 0.17 with the highest and lowest Sinhala-script checkpoints dropped. Every fit is reliably below the 1.0 of full transfer ($p = 8.6 \times 10^{-7}$ for the checkpoint-level form) and above the 0.0 of total flattening (Appendix F).

On safety, accuracy hides it. Accuracy on SOLD falls by 1.6 to 11.3 points against a 59.4% majority baseline, which looks survivable, while the median checkpoint keeps only 45% of its Matthews correlation. Five of the six checkpoints competent on SinhalaMMLU lose significantly; the SOLD screen itself admits five of the eleven, with Hormoz-8B falling just outside it on a 90.2% single-label rate, and Appendix E reports both cohorts. The mechanism is a retreat toward triviality, and Qwen2-7B-Instruct more than doubles its rate of predicting the offensive class, 10.5% to 23.4%, without becoming more accurate. A deployment tracking accuracy would see a five point wobble while over half the usable signal disappeared.

The effect is neither a tokenization artifact nor output degeneration: all eleven checkpoints receive *fewer* tokens in the Romanized condition on all three tasks, a median 2.09 per word against 9.5, and the invalid rate never exceeds 0.03% in either condition (Appendix E). The models keep answering and are simply wrong.

5 Implications for evaluation

Five practices follow from the measurements above, each traceable to one of them.

Normalize by a unit both conditions share. Perplexity conflates capability with tokenizer fertility, and bits per byte removes the tokenizer only to reintroduce the encoding, since a byte buys less text in Sinhala than in Latin (§3). Normalize by a unit whose count is invariant across the pair, words being the obvious one, and report the denominator with the number.

Evaluate in deployed registers. Formal-script-only evaluation overstates practical capability wherever a Romanized variant dominates use. Benchmarks should test Romanized input directly, for the drop and for the compression (§4).

Screen for native competence. Averaging in checkpoints that sit near chance in the formal script hides degradation by construction. Report the guessing baseline and condition group averages on demonstrated competence (§4).

Select metrics robust to class imbalance. Under imbalance, accuracy stays anchored to the majority baseline while discriminative power collapses. Skewed tasks want chance-adjusted metrics, Matthews correlation or macro-F1 (§4).

Validate in the target register. Formal-script rankings do not predict Romanized rankings, and the stronger the model the more it loses. Select checkpoints inside the deployment register, transliterating before inference where feasible.

6 Limitations

The Romanized side of the downstream items is produced by our transliterator rather than typed by users, and that is the limitation to weigh first, because no parallel resource exists at that scale. Rather than assume it is neutral we measured it: our text is 0.17 to 0.24 bits per byte *harder* than what native speakers typed for the same sentences, so our downstream gaps are closer to upper bounds than to lower bounds. That control covers six checkpoints, none of them among those carrying the downstream result, which a few hundred human-typed items would fix (§H). The intrinsic results do not share the limitation, since their Romanized side was human-typed, and they show the same flattening. Everything here is also one language, and the floor effect we describe may not transfer, although Khullar et al. [4] report compatible magnitudes for six South Asian languages. We cannot exclude Sinhala-script evaluation material in pretraining, and everything is zero-shot on open weights up to 14.7B, so this is what a user gets out of the box and not a ceiling. Appendix I has the rest.

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A Related work

Sinhala and low-resource evaluation. Sinhala is low-resource and its tooling has grown in a fragmented way [1, 16, 17]. Recent resources are almost entirely Sinhala-script: SinhalaMMLU [10], SOLD [2], the Sinhala split of Global PIQA [11, 12] and encoder work such as Dhananjaya et al. [18]. The only benchmark treating script as a variable is Rajapakse and Weerasinghe [5], which we extend.

Romanization gaps elsewhere. Romanized South Asian text has been studied mainly as a resource and identification problem [19, 20]. J et al. [21] show romanization can *help* when models are adapted for it, the mirror image of our zero-shot setting. The closest same-language work trains a model for the register: Chamikara and Sumanathilaka [22] fine-tune multilingual encoders for hate speech detection on Romanized Sinhala, and Hewapathirana and Sumanathilaka [23] build a classifier over Romanized Sinhala social media. We instead ask what general-purpose generative models do out of the box. Closest in spirit is Khullar et al. [4], who find weighted F1 falling by as much as 24 points on Romanized clinical queries. We add a paired item-level design with a competence screen, a tokenizer-independent account of the same models, and a test of whether the gap is a tokenization artifact.

Why not perplexity. Bits per byte exists so that models with different vocabularies can be compared [6], and Paloma adopts it for that reason [7]. Tokenizers treat languages very unequally [8, 9] and over-segmentation costs accuracy on low-resource tasks [24]. To our knowledge ours is the first work to show that the substantive conclusions of a published benchmark change when the normalizer changes.

B Setup details

Hardware and compute. The full study is roughly 80 hours of GPU wall clock. The intrinsic sweep is about 10 hours over 46,500 scored sequences, and the downstream run about 70 hours over 208,538 generations, of which roughly 25 hours are Phi-4-14B alone. Inference is fp16 on two T4 GPUs, except Phi-4-14B which ran on a single 16 GB RTX 4070 Ti SUPER. The transliteration control of §H adds 9,000 sequences in fp32 on 8 CPU cores in under an hour. No training was performed at any point.

Prompting. All prompts are zero-shot, greedily decoded with a 40-token limit, and answers are extracted by pattern match with anything unparseable scored as wrong. We chose one template from three candidates using a pilot of 3,240 generations over nine checkpoints, both script conditions and 20 items per dataset (Table 1). The direct template won on SinhalaMMLU, the largest set, and was the only candidate there that never produced unparseable output. An answer-first template scored 1.4 points higher on SOLD and 7.2 higher on Global PIQA, and we applied the direct template everywhere anyway, because a template chosen per dataset would confound task with prompt, and because 20 items cannot resolve differences of that size. What a paired comparison needs is not an optimal template but one that does not favor a script, and the evidence for that is the main run, where the unparseable rate never exceeds 0.03% in either condition for the competent cohort (Table 10). Twenty of the pilot’s SinhalaMMLU items, 0.3% of the benchmark, are also in the final evaluation set.

Statistics. Both conditions score the same items, so we use McNemar’s exact or continuity-corrected test [14, 25] and report discordant pair counts. Within each dataset we control the family-wise error rate over checkpoints with Holm’s procedure [15]. Intervals on gaps, macro-F1 differences, Matthews correlation differences and regression slopes come from 10,000 paired bootstrap resamples over items, except for the cross-checkpoint regressions of §4, which resample checkpoints. For SOLD we lead with Matthews correlation [26, 27] because the classes are imbalanced and accuracy sits close to the majority baseline. Unparseable output is scored as wrong for accuracy. In the Matthews correlation it enters as a prediction of the non-offensive class rather than symmetrically, so it is not penalised identically in both directions; every checkpoint carrying a SOLD result produced zero unparseable output in both conditions, so the choice affects nothing we report. Item-level effects are also fitted as logistic regressions with checkpoint fixed effects and cluster-robust standard errors by

Table 1: The prompt-template pilot: 9 checkpoints, 3 templates, 2 script conditions, 20 items per dataset, 3,240 generations. Accuracy counts unparseable output as wrong. T1 was applied to all three datasets even though T3 scored higher on SOLD and Global PIQA, so that no dataset gets a different prompting regime. At 20 items per dataset none of these differences is resolvable, which is why the pilot is treated as a format check.

Template	Accuracy %			Invalid %	
	S	R	both	S	R
<i>SinhalaMMLU</i>					
T1 direct (used)	28.3	25.0	26.7	0.0	0.0
T2 two-shot	18.3	28.9	23.6	10.6	2.8
T3 answer first	27.2	23.9	25.6	0.6	0.6
<i>SOLD</i>					
T1 direct (used)	51.1	51.1	51.1	10.6	8.9
T2 two-shot	42.8	44.4	43.6	10.6	5.6
T3 answer first	49.4	55.6	52.5	11.1	8.9
<i>Global PIQA</i>					
T1 direct (used)	48.9	40.6	44.7	4.4	3.3
T2 two-shot	46.1	48.3	47.2	13.9	0.6
T3 answer first	52.2	51.7	51.9	3.9	5.6

item, which put the odds ratio for Romanized input on SinhalaMMLU at 0.70 ($p < 10^{-33}$) and give a significant difficulty interaction ($p < 0.001$).

C Reproducing the published benchmark

Table 3 sets our recomputation beside every headline number of Rajapakse and Weerasinghe [5] on its own 24 checkpoints. Per-checkpoint absolute deviations from the published perplexities have a median of 0.05% on Sinhala-script text and 0.004% on Romanized text, and 0.39% on mixed-script text once two outliers are excluded, Phi-4-14B and Mistral-Nemo-Base-2407. Everything in §3 is therefore the metric and not a different run. Table 4 applies Equation 1 per checkpoint, and Figure 3 shows the same split as stacked bars. Figure 2 shows the disagreement between the two metrics directly.

On aggregating the split. Equation 1 is exact for each checkpoint, but its two terms are not medians that compose. On the published 24 the median total is 8.29 bits while the median normalizer and per-byte-loss terms are 4.37 and 4.56, which sum to 8.93 rather than 8.29 because no single checkpoint attains all three medians. Read multiplicatively those two terms are 21-fold and 24-fold, and 21×24 is 504, not 312. An earlier version of this paper quoted that pair; the quantity that does aggregate honestly is the normalizer’s per-checkpoint *share*, reported in Table 4 and summarised in §3. Note also that the 24-fold figure is a per-token quantity, since the loss term is bytes per token times a change in bits per byte; the per-byte figure is 2.16 bits per byte, a factor of 4.5.

Why we lead with matched units. Bits per byte removes the tokenizer from the denominator but not the encoding, because $\text{BPB}_r/\text{BPB}_u = (\text{NLL}_r/\text{NLL}_u) \times (B_u/B_r)$ exactly, and the second factor has a median of 2.30 here for reasons that have nothing to do with any model. Whitespace word counts are identical within all 31 pairs, so bits per word carries no such factor and its ratio is the ratio of total loss on identical content: a median 1.02 on the published 24 and 1.06 over all 31, with 12 checkpoints cheaper on Romanized text in both pools. Under the word unit the normalizer share of the reported gap is 99% on the 24 and 94% over all 31.

Two further details are worth stating. Over all 31 checkpoints the normalizer share of the total falls to 39%, and for the six Qwen3.5 and Gemma checkpoints the normalizer term is slightly negative, because their tokenizers spend about the same number of bytes per token in both scripts. Those six also hold the five smallest ratios in the pool, with the four Qwen3.5 entries between 58-fold and 64-fold against a pool median of 294-fold, so the eye-catching part of a 312-fold figure is concentrated in the models whose tokenizers handle Sinhala worst. How completely the two metrics come apart also depends on the pool: on the 24 published checkpoints the rank correlation between perplexity

Table 2: All evaluated checkpoints, with Hugging Face identifiers and tokenizer fertility on the parallel corpus. ‡ marks the eleven used downstream and • the 24 that Rajapakse and Weerasinghe [5] evaluated. Sinhala-script fertility varies by more than a factor of three across tokenizers, whereas Romanized fertility is nearly constant.

Checkpoint	B	Hugging Face identifier	tokens / word	
			Sinh.	Rom.
OPT-350M•	0.35	facebook/opt-350m	14.1	2.4
Pythia-410M•	0.41	EleutherAI/pythia-410m	14.0	2.2
BLOOM-560M•	0.56	bigscience/bloom-560m	8.9	1.9
BLOOM-1B1•	1.10	bigscience/bloom-1b1	8.9	1.9
TinyLlama-1.1B-Chat‡•	1.10	TinyLlama/TinyLlama-1.1B-Chat-v1.0	9.8	2.4
Llama-3.2-1B•	1.20	meta-llama/Llama-3.2-1B	9.7	2.2
Cerebras-GPT-1.3B•	1.30	cerebras/Cerebras-GPT-1.3B	14.0	2.3
OPT-1.3B•	1.30	facebook/opt-1.3b	14.1	2.4
LaMini-GPT-1.5B‡•	1.50	MBZUAI/LaMini-GPT-1.5B	14.0	2.3
Qwen1.5-1.8B•	1.80	Qwen/Qwen1.5-1.8B	7.6	2.1
OPT-2.7B•	2.70	facebook/opt-2.7b	14.1	2.4
StableLM-Zephyr-3B‡•	2.80	stabilityai/stablelm-zephyr-3b	14.0	2.2
BLOOM-3B•	3.00	bigscience/bloom-3b	8.9	1.9
SmolLM3-3B‡•	3.10	HuggingFaceTB/SmolLM3-3B	9.5	2.1
Llama-3.2-3B•	3.20	meta-llama/Llama-3.2-3B	9.7	2.2
Qwen3.5-4B-Base	4.00	Qwen/Qwen3.5-4B-Base	4.3	2.0
Qwen3.5-4B‡	4.00	Qwen/Qwen3.5-4B	4.3	2.0
Mistral-7B-v0.3•	7.20	mistralai/Mistral-7B-v0.3	9.3	2.4
Zephyr-7B-beta‡•	7.20	HuggingFaceH4/zephyr-7b-beta	9.3	2.4
Qwen2-7B•	7.60	Qwen/Qwen2-7B	7.6	2.1
Qwen2-7B-Instruct‡	7.60	Qwen/Qwen2-7B-Instruct	7.6	2.1
Llama-3.1-8B•	8.00	meta-llama/Meta-Llama-3.1-8B	9.7	2.2
Llama-3.1-8B-Instruct‡	8.00	meta-llama/Llama-3.1-8B-Instruct	9.7	2.2
Hormoz-8B‡•	8.00	mann-e/Hormoz-8B	8.6	2.1
Minitron-8B-Base•	8.40	nvidia/Minitron-8B-Base	14.2	1.9
Gemma-7B•	8.50	google/gemma-7b	4.5	2.0
Qwen3.5-9B-Base	9.00	Qwen/Qwen3.5-9B-Base	4.3	2.0
Qwen3.5-9B‡	9.00	Qwen/Qwen3.5-9B	4.3	2.0
Gemma-2-9B	9.20	google/gemma-2-9b	4.5	2.0
Mistral-Nemo-Base-2407•	12.20	mistralai/Mistral-Nemo-Base-2407	14.2	2.2
Phi-4-14B‡•	14.70	microsoft/phi-4	9.5	2.1

and bits per byte is moderate ($\rho = 0.46$, $p = 0.02$), still leaving four fifths of bits-per-byte rank variance unexplained, and adding the seven newer checkpoints removes even that.

D Full intrinsic results

Table 5 gives all 31 checkpoints, sorted by Sinhala-script bits per byte. The two orderings disagree almost completely. Bits per word is the sharpest cross-script unit, because a sentence and its romanization contain the same number of words while Sinhala UTF-8 spends 2.65 bytes per character against 1.00 for Latin.

The mixed-script condition behaves quite differently from the Romanized one and reproduces the earlier finding that script handling tracks general ability. Sinhala-script and mixed-script bits per byte are strongly rank-correlated ($\rho = 0.94$) at a median gap of only 0.49 bits, against $\rho = 0.41$ for Romanized. Interleaving scripts is a mild perturbation. Replacing one entirely is not.

Table 6 gives the character n -gram reference points, fit on the same corpus under five-fold cross-validation so every score is out of sample, and Figure 4 plots them against the checkpoints. On Romanized Sinhala the best n -gram reaches 2.64 bits per byte and a character bigram 3.06, while the best checkpoint reaches 3.29 and the median 3.81, so no checkpoint beats the bigram. On the Sinhala script the picture is ordinary, with the character trigram at 1.16 and one checkpoint below it. The n -gram is a cheap in-domain model and the comparison favors it in that respect, so we read this as Romanized-condition checkpoints failing to exploit structure beyond what a character bigram captures, rather than as a claim about what they know. Two BLOOM checkpoints sit near the bottom

Table 3: Our recomputation against the published benchmark, on its own 24 checkpoints. Deviations are per-checkpoint absolute percentage differences from the published perplexities.

Quantity	Published	Ours
median PPL ratio, Sinh. to Rom.	312.3	312.3
its range	149.1 to 769.7	148.5 to 769.3
median PPL ratio, Sinh. to mixed	2.0	2.0
$\rho(\text{params, Sinh. PPL})$	+0.09	-0.05
$\rho(\text{params, Rom. PPL})$	-0.09	+0.11
$\rho(\text{Sinh. PPL, Rom. PPL})$	+0.52	+0.53
$\rho(\text{Sinh. PPL, mixed PPL})$	+0.96	+0.98
CV, Sinh. PPL	66.4%	67.8%
CV, Rom. PPL	62.3%	63.8%
median dev. , Sinh. PPL	—	0.047%
median dev. , Rom. PPL	—	0.004%

Table 4: Equation 1 applied to every checkpoint, sorted by the reported perplexity ratio. “ratio” is Romanized over Sinhala-script perplexity and “bits” its base-2 logarithm, which the two terms partition exactly. The last column is the normalizer’s share of the total, and ● marks the 24 checkpoints of the published pool.

Checkpoint	ratio	bits	token-count normalizer		per-byte loss		norm. %
			bits	factor	bits	factor	
Qwen3.5-9B	58	5.86	-0.34	0.8	6.20	73.6	-6
Qwen3.5-9B-Base	58	5.87	-0.34	0.8	6.21	74.0	-6
Qwen3.5-4B-Base	63	5.98	-0.35	0.8	6.33	80.6	-6
Qwen3.5-4B	64	6.01	-0.36	0.8	6.36	82.4	-6
Gemma-7B●	149	7.21	-0.20	0.9	7.42	171.1	-3
Mistral-7B-v0.3●	175	7.45	2.72	6.6	4.73	26.6	37
Zephyr-7B-beta●	183	7.52	3.08	8.4	4.44	21.7	41
Qwen1.5-1.8B●	196	7.62	2.83	7.1	4.79	27.6	37
Qwen2-7B●	216	7.75	2.68	6.4	5.07	33.6	35
OPT-350M●	226	7.82	4.74	26.6	3.09	8.5	61
TinyLlama-1.1B-Chat●	232	7.86	2.95	7.7	4.90	29.9	38
LaMini-GPT-1.5B●	244	7.93	5.27	38.5	2.67	6.4	66
Qwen2-7B-Instruct	249	7.96	2.74	6.7	5.22	37.3	34
Phi-4-14B●	287	8.16	3.49	11.2	4.68	25.6	43
OPT-1.3B●	290	8.18	4.53	23.1	3.65	12.6	55
OPT-2.7B●	294	8.20	4.42	21.4	3.78	13.7	54
Llama-3.1-8B●	301	8.24	3.07	8.4	5.16	35.8	37
BLOOM-1B1●	323	8.34	4.72	26.3	3.62	12.3	57
Llama-3.1-8B-Instruct	325	8.35	3.28	9.7	5.07	33.5	39
Gemma-2-9B	357	8.48	-0.20	0.9	8.68	411.2	-2
Cerebras-GPT-1.3B●	364	8.51	4.39	21.0	4.11	17.3	52
BLOOM-560M●	389	8.60	4.93	30.6	3.67	12.7	57
Llama-3.2-3B●	417	8.70	3.33	10.1	5.37	41.4	38
Llama-3.2-1B●	419	8.71	3.42	10.7	5.29	39.0	39
StableLM-Zephyr-3B●	466	8.87	5.41	42.6	3.45	10.9	61
BLOOM-3B●	469	8.87	4.35	20.4	4.53	23.0	49
Pythia-410M●	482	8.91	4.45	21.9	4.46	22.0	50
Mistral-Nemo-Base-2407●	691	9.43	4.85	28.8	4.58	24.0	51
Hormoz-8B●	698	9.45	3.72	13.2	5.72	52.9	39
Minitron-8B-Base●	710	9.47	5.56	47.1	3.91	15.1	59
SmolLM3-3B●	769	9.59	4.50	22.6	5.09	34.0	47

of Table 5 on Sinhala-script text while being unremarkable on Romanized text, and BLOOM is the only family in the pool with documented Sinhala pretraining exposure [28].

E Full downstream results

Table 7 gives all eleven checkpoints on SinhalaMMLU and SOLD, with macro-F1 alongside Matthews correlation. Table 8 breaks SinhalaMMLU into difficulty and domain strata pooled over the six competent checkpoints, Table 9 gives Global PIQA per checkpoint, and Table 10 gives the two controls of §4 per checkpoint, the unparseable rates and the ratio of Romanized to Sinhala-script input tokens. Figure 5 shows the SOLD result, where accuracy and Matthews correlation disagree about whether anything happened.

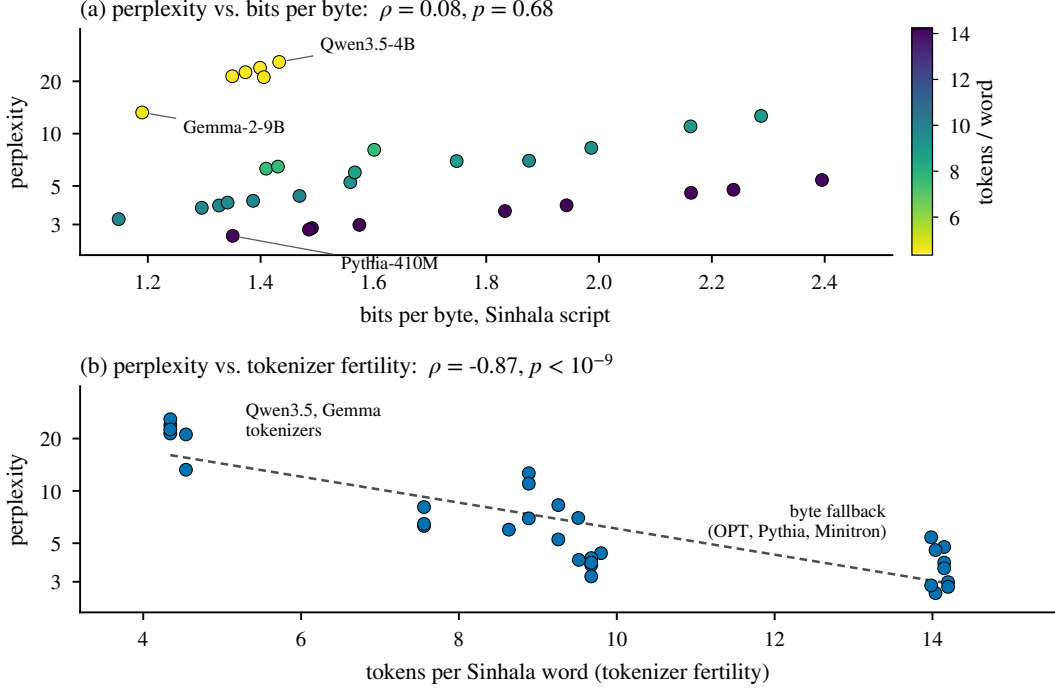


Figure 2: (a) Sinhala-script perplexity against Sinhala-script bits per byte over 31 checkpoints. The two are unrelated ($\rho = 0.08, p = 0.68$). (b) Perplexity against tokenizer fertility, which explains it almost fully ($\rho = -0.87, p < 10^{-9}$).

Two numbers in Table 7 are worth flagging. LaMini-GPT-1.5B and TinyLlama-1.1B-Chat appear to improve under romanization, and both are artifacts of their failing to produce parseable output on the Sinhala-script side rather than script effects, which is what the competence screen exists to catch. The SOLD offensive-class rates behind the retreat toward triviality are 10.5% to 23.4% for Qwen2-7B-Instruct and 9.8% to 20.7% for Hormoz-8B.

Which cohort is which. Three sets recur and we name them here once. Six checkpoints pass the screen on SinhalaMMLU: Qwen3.5-4B, Qwen2-7B-Instruct, Llama-3.1-8B-Instruct, Hormoz-8B, Qwen3.5-9B and Phi-4-14B. Five pass on SOLD, the same set without Hormoz-8B, which clears the significance test but places 90.2% of its labels on one class against a 90% ceiling. The SinhalaMMLU pooled gap and the strata of Table 8 use the six; the SOLD pooled gap and the attestation terciles of Table 17 use the five; the Matthews-correlation results of §4 are stated over the SinhalaMMLU six so that the same checkpoints carry both tasks. The flattening regressions of Table 11 use neither screen, taking all ten checkpoints that produce parseable output, and are refitted on the competent six there.

Sensitivity of the screen. Both thresholds are free parameters, so we swept them. On SinhalaMMLU the six are unchanged at $p < 0.001$ and $p < 0.01$ and under either an 80% or a 90% concentration rule; $p < 0.05$ admits Zephyr-7B-beta at $p = 0.045$, and a 95% rule admits SmolLM3-3B at a 93.3% single-option rate. SOLD is the sensitive task, turning on Hormoz-8B’s two tenths of a point. Global PIQA admits nobody under any setting we tried. We report the screen as specified rather than tuned, and the SinhalaMMLU cohort that carries the headline gap is the same under three of the four settings.

F Every refit of the flattening regressions

Table 11 lists every fit. One deserves explanation, because it is the natural way to state the result and it is a trap. Writing headroom as Sinhala-script accuracy minus chance, the paired loss fits headroom at loss = $0.78 \times \text{headroom} - 1.06$ with $R^2 = 0.96$. But Sinhala-script accuracy appears on both axes, so a high fit is close to guaranteed. Shuffling the Romanized scores to destroy any association still

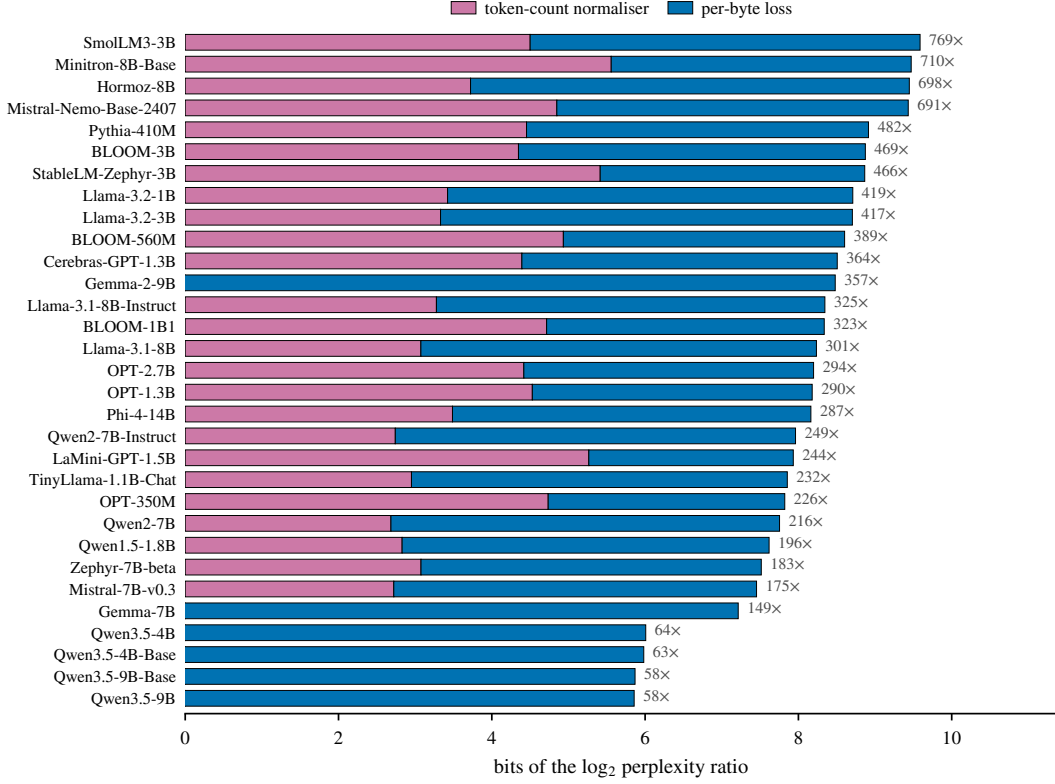


Figure 3: The split of Equation 1 per checkpoint. The two terms sum exactly to the reported gap in bits.

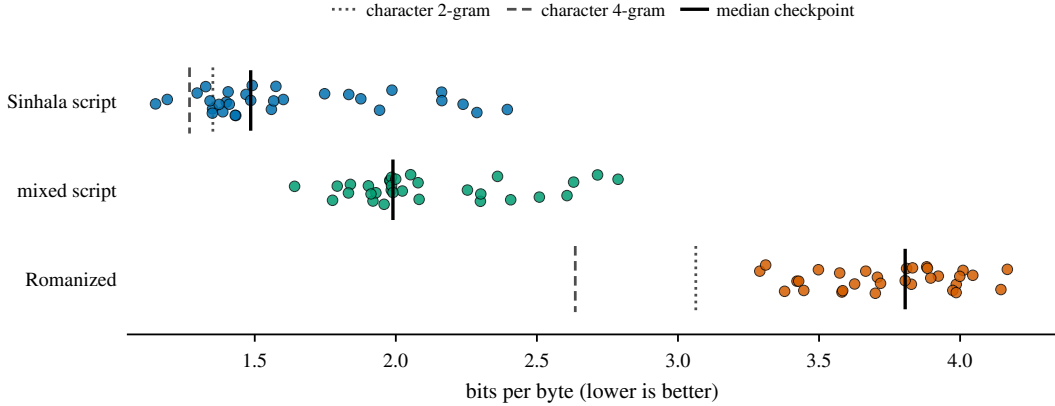


Figure 4: Checkpoint bits per byte against character n -gram references in both conditions. On Romanized Sinhala every checkpoint sits above the bigram.

yields a mean slope of 1.00 and a mean R^2 of 0.94, with 8.9% of permutations reaching $R^2 \geq 0.96$. We do not treat that fit as evidence, and we report it here because the same construction is easy to reach for in any robustness study. The levels from we do rely on returns a mean slope of 0.00 and a mean R^2 of 0.11 under the same null, and only 0.5% of permutations reach the observed fit. The cell-level refit is 0.48 when cells are weighted by item count. The two views are complementary rather than independent, since they aggregate the same generations differently, but they agree on the shape and disagree usefully on the amount: content-level transfer is about twice model-level transfer.

How far the checkpoint-level slope moves. Because the two estimates differ by a factor of two, it is fair to ask what the smaller one is sensitive to. Weighting checkpoints by item count does nothing,

Table 5: All 31 checkpoints, sorted by Sinhala-script bits per byte. ‡ marks the eleven evaluated downstream. The perplexity and bits-per-byte orderings disagree almost completely: Gemma-2-9B is 2nd by bits per byte and 26th by perplexity, and Qwen3.5-9B-Base moves from 28th to 6th.

Checkpoint	B	Sinhala script			Romanized			Mixed	
		PPL	BPB	BPW	PPL	BPB	BPW	PPL	BPB
Llama-3.1-8B	8.0	3.22	1.148	16.3	971	3.584	22.1	6.36	1.641
Gemma-2-9B	9.2	13.23	1.190	16.9	4726	3.999	24.7	48.99	1.792
Llama-3.2-3B	3.2	3.74	1.296	18.4	1561	3.831	23.6	7.40	1.776
Llama-3.1-8B-Instruct‡	8.0	3.86	1.326	18.9	1256	3.718	22.9	7.94	1.839
Phi-4-14B‡	14.7	4.01	1.341	19.1	1149	3.446	21.3	8.08	1.832
Qwen3.5-9B-Base	9.0	21.39	1.350	19.2	1250	3.290	20.3	78.56	1.958
Pythia-410M	0.4	2.58	1.351	19.2	1245	3.707	22.9	4.91	1.978
Qwen3.5-9B‡	9.0	22.56	1.373	19.5	1308	3.311	20.4	84.24	1.990
Llama-3.2-1B	1.2	4.11	1.387	19.7	1719	3.881	24.0	8.54	1.903
Qwen3.5-4B-Base	4.0	23.92	1.399	19.9	1513	3.378	20.8	83.32	1.985
Gemma-7B	8.5	21.13	1.406	20.0	3139	3.805	23.5	66.00	1.930
Qwen2-7B	7.6	6.29	1.410	20.0	1357	3.498	21.6	14.11	1.919
Qwen2-7B-Instruct‡	7.6	6.46	1.431	20.3	1611	3.581	22.1	15.45	1.985
Qwen3.5-4B‡	4.0	25.84	1.433	20.4	1663	3.422	21.1	90.71	2.023
TinyLlama-1.1B-Chat‡	1.1	4.38	1.469	20.9	1014	3.923	24.2	8.92	1.980
Mistral-Nemo-Base-2407	12.2	2.80	1.486	21.1	1938	3.896	24.0	4.63	1.912
Cerebras-GPT-1.3B	1.3	2.86	1.491	21.2	1040	3.700	22.8	5.23	2.052
Mistral-7B-v0.3	7.2	5.26	1.559	22.2	921	3.884	24.0	10.15	1.984
Hormoz-8B‡	8.0	5.98	1.567	22.3	4176	4.045	25.0	13.58	2.083
Minitron-8B-Base	8.4	2.98	1.575	22.4	2119	3.428	21.2	5.08	2.000
Qwen1.5-1.8B	1.8	8.07	1.601	22.8	1585	3.573	22.1	17.60	2.079
BLOOM-3B	3.0	6.95	1.748	24.8	3258	3.626	22.4	15.90	2.254
OPT-2.7B	2.7	3.59	1.833	26.1	1053	3.974	24.5	6.26	2.300
SmolLM3-3B‡	3.1	6.98	1.876	26.7	5373	4.145	25.6	13.89	2.301
OPT-1.3B	1.3	3.87	1.942	27.6	1122	4.010	24.8	6.82	2.407
Zephyr-7B-beta‡	7.2	8.28	1.986	28.2	1516	4.167	25.7	15.76	2.360
BLOOM-1B1	1.1	11.00	2.162	30.7	3558	3.666	22.6	24.52	2.607
StableLM-Zephyr-3B‡	2.8	4.57	2.163	30.8	2129	3.986	24.6	7.51	2.509
OPT-350M	0.3	4.75	2.238	31.8	1076	3.986	24.6	8.15	2.630
BLOOM-560M	0.6	12.64	2.287	32.5	4915	3.810	23.5	28.01	2.715
LaMini-GPT-1.5B‡	1.5	5.41	2.395	34.1	1322	3.827	23.6	9.46	2.788

Table 6: Held-out add-1 character n -gram models against the 31 checkpoints, five-fold cross-validated so every score is out of sample. “# LMs <” counts checkpoints achieving lower bits per byte than the n -gram model.

Model	Sinhala script		Romanized	
	BPB	# LMs <	BPB	# LMs <
character 1-gram	1.815	22	3.977	24
character 2-gram	1.352	7	3.063	0
character 3-gram	1.162	1	2.685	0
character 4-gram	1.269	2	2.636	0
character 5-gram	1.474	15	2.947	0
character 6-gram	1.680	21	3.381	3
best checkpoint	1.148	—	3.290	—
median checkpoint	1.486	—	3.805	—

since all ten score the same 6,879 items and the weights are equal; this is why only the cell-level fit has a weighted variant. Restricting to the six checkpoints that pass the competence screen lowers the slope to 0.12, and dropping the highest and lowest Sinhala-script checkpoints lowers it to 0.17. Both move away from the cell-level 0.45 rather than toward it, so the gap between the two views is not an artifact of either cohort or of the extremes, and we lead with the cell-level figure because it is the more conservative of the two rather than because it is the more robust. Neither fit treats checkpoints from one family as correlated, which they are: three of the ten are Qwen checkpoints, and with $n = 10$ a one-checkpoint-per-family refit leaves too few points to interpret. The bootstrap intervals should be read with that in mind.

Table 7: All eleven downstream checkpoints, ordered by size. S is the Sinhala script, R is Romanized, Δ is the paired difference in points, p_{Holm} is McNemar’s test corrected across checkpoints within a task, and MCC is Matthews correlation with a paired-bootstrap p . † marks checkpoints passing the competence screen.

Checkpoint	SinhalaMMLU (6,879 items)						SOLD (2,500 items)					
	Acc. %					p_{Holm}	macro-F1		MCC		p_{Holm}	
	B	S	R	Δ	95% CI		S	R	S	R		
TinyLlama-1.1B-Chat	1.1	21.7	21.9	-0.2	[-0.5, 0.1]	.8273	6.4	24.2	-0.019	-0.043	1.0000	
LaMini-GPT-1.5B	1.5	0.9	18.8	-17.9	[-18.8, -16.9]	< .0001	5.1	28.3	0.000	0.024	1.0000	
StableLM-Zephyr-3B	2.8	22.9	24.0	-1.1	[-2.3, 0.1]	.4275	38.6	38.9	0.036	0.001	.9408	
SmolLM3-3B	3.1	25.1	25.5	-0.3	[-1.0, 0.4]	.8273	29.7	29.9	0.025	0.017	1.0000	
Qwen3.5-4B†	4.0	36.7	26.1	+10.6	[9.3, 12.0]	< .0001	60.2	55.1	0.219	0.127	.0018	
Zephyr-7B-beta	7.2	24.3	24.9	-0.6	[-1.7, 0.5]	.8273	39.7	39.6	0.060	0.060	1.0000	
Qwen2-7B-Instruct†	7.6	29.0	27.0	+2.0	[0.8, 3.2]	.0055	49.9	50.3	0.148	0.042	.0018	
Llama-3.1-8B-Instruct†	8.0	32.5	27.1	+5.4	[4.1, 6.7]	< .0001	60.1	52.6	0.259	0.105	< .0001	
Hormoz-8B†	8.0	26.0	25.2	+0.8	[-0.5, 2.0]	.8273	47.9	51.9	0.108	0.090	1.0000	
Qwen3.5-9B†	9.0	44.1	28.7	+15.5	[14.0, 16.9]	< .0001	63.1	52.0	0.280	0.069	< .0001	
Phi-4-14B†	14.7	28.7	27.2	+1.5	[0.2, 2.8]	.1286	54.4	50.8	0.152	0.076	.0084	

Table 8: SinhalaMMLU by stratum, pooled over the six competent checkpoints. Item counts are per checkpoint and domains are ordered by loss.

Stratum	Items	S %	R %	Δ	p_{Holm}
<i>by difficulty</i>					
Easy	1,851	37.0	29.2	7.9	< .0001
Medium	2,503	34.9	28.7	6.2	< .0001
Hard	2,525	27.7	23.4	4.3	< .0001
<i>by domain</i>					
Social Science	1,059	37.2	26.9	10.3	< .0001
Business Studies	463	34.4	27.1	7.3	< .0001
Other	1,014	31.7	24.6	7.0	< .0001
Stem	614	33.3	26.5	6.7	< .0001
Language	388	29.0	24.4	4.6	< .0001
Humanities	3,341	31.9	27.9	4.1	< .0001

G Do intrinsic metrics track downstream fragility?

For the ten checkpoints measured both ways, absolute Sinhala-script bits per byte tracks the Sinhala-MMLU gap ($\rho = -0.83$, 95% CI -1.00 to -0.30 , leave-one-out -0.90 to -0.77) and the loss of Matthews correlation on SOLD ($\rho = -0.78$, CI -1.00 to -0.36 , leave-one-out -0.87 to -0.73), which is the flattening of §4 seen from the intrinsic side. The negative result is the more useful one. The bits-per-byte *gap*, the natural analogue of the degradation ratio prior work reports, shows no association ($\rho = -0.15$, CI -0.89 to $+0.63$), and neither does the perplexity ratio ($\rho = -0.52$). For the gap even the leave-one-out range straddles zero, so we cannot sign it.

All of this is exploratory: ten heterogeneous checkpoints, no held-out validation, no pre-registration, and intervals wide enough to reach -1.00 . The levels-versus-gaps contrast is stable under leave-one-out, but Table 12 is a hypothesis for a larger checkpoint set rather than a screening instrument.

H The transliterator, and what it costs

Our transliterator is a deterministic grapheme-level converter that parses Sinhala as an abugida and uses the digraph conventions speakers type rather than a scholarly scheme, so the dental stop becomes *th* and the labiodental approximant becomes *w*. It runs locally, covers every input, and collapses only the consonant distinctions casual Singlish also collapses.

Against human references. We validated it on 755,000 reference items from the Swa-bhasha Resource Hub [13]: 4,397 authentic social-media sentence pairs, 450,587 words each carrying several accepted spellings, and a 300,000-sentence machine-augmented cross-check. About 455,000 of the

Table 9: Global PIQA, 100 items. “disc. b/c” are the discordant pair counts entering McNemar’s test. No checkpoint is above chance in the Sinhala-script condition and no comparison survives correction, so we treat this task as underpowered rather than as corroboration.

Checkpoint	Acc. %		Δ	disc.	
	S	R		b/c	p_{Holm}
TinyLlama-1.1B-Chat	40	33	+7	12/5	1.0000
LaMini-GPT-1.5B	0	36	-36	0/36	< .0001
StableLM-Zephyr-3B	45	44	+1	15/14	1.0000
SmolLM3-3B	53	51	+2	14/12	1.0000
Qwen3.5-4B	57	49	+8	22/14	1.0000
Zephyr-7B-beta	48	50	-2	18/20	1.0000
Qwen2-7B-Instruct	45	42	+3	21/18	1.0000
Llama-3.1-8B-Instruct	53	48	+5	22/17	1.0000
Hormoz-8B	47	48	-1	13/14	1.0000
Qwen3.5-9B	53	45	+8	25/17	1.0000
Phi-4-14B	46	46	+0	18/18	1.0000

Table 10: Unparseable-output rates, and τ , the ratio of Romanized to Sinhala-script input tokens on SinhalaMMLU. τ is below one for every checkpoint, so the Romanized condition is the shorter input in every case.

Checkpoint	MMLU inv. %		PIQA inv. %		τ
	S	R	S	R	
TinyLlama-1.1B-Chat	1.47	1.03	18	28	0.45
LaMini-GPT-1.5B	95.93	16.35	100	28	0.31
StableLM-Zephyr-3B	3.82	2.63	0	0	0.32
SmolLM3-3B	0.00	0.00	0	0	0.48
Qwen3.5-4B	0.00	0.00	0	0	0.72
Zephyr-7B-beta	2.11	0.64	0	0	0.49
Qwen2-7B-Instruct	0.03	0.01	0	0	0.51
Llama-3.1-8B-Instruct	0.00	0.00	0	0	0.45
Hormoz-8B	0.00	0.00	0	0	0.50
Qwen3.5-9B	0.00	0.00	0	0	0.72
Phi-4-14B	0.00	0.03	0	0	0.42

items are therefore human-supplied and the remainder machine-augmented, and we report the three corpora separately for that reason. Our mapping tables were written and frozen before this corpus entered the project and were never revised against it, so no part of the comparison is fitted to what it scores. One comparator is not independent in that way and we flag it where it appears: the *nisansa_w* variant applies a single orthographic rewrite chosen from a statistic measured on this same corpus, and both its rows are reported side by side in Table 13. Scored against the closest accepted reference by character error rate, chrF [29] and exact match, ours has the lowest error rate of four methods on all three corpora, 0.182, 0.121 and 0.112, and a paired Wilcoxon test separates it from Aksharamukha [30], uroman [31] and the web romanizer of de Silva and Ranathunga [12] at $p < 10^{-90}$ on the two large corpora (Table 13). The residual difference from human typing is one convention rather than phonemic error. All four methods double long vowels far more often than people do, 0.81 per word against 0.12. Word by word against the human-typed side of the parallel corpus, 44.6% of our forms match exactly and a further 27.6% match once that convention is normalized, leaving 27.8% residual, and against the multi-reference lexicon of 449,598 words and 7,075,844 spellings, 88.6% of our spellings are ones a human actually supplied for that word, against 84.2% for the native speakers who typed the same sentences (Tables 14 and 15). That our output scores *above* the humans on this measure is a reason to read it narrowly: attestation says our spellings are ones people use, not that our text reads as a person would have typed it. Naturalness in that stronger sense we do not measure, and the bits-per-byte control below is the closer test.

What the substitution costs a model. The Romanized side of the downstream items is ours rather than human-typed, so we measured what that substitution costs. On the 500 parallel sentences, whose

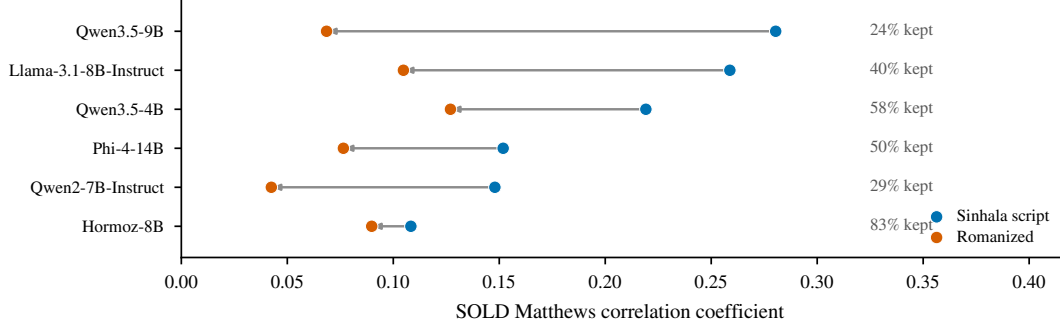


Figure 5: SOLD under romanization. Accuracy moves a few points against a 59.4% majority baseline while Matthews correlation loses a median 55% of its value.

Table 11: Every refit of the two flattening regressions. The levels form regresses Romanized on Sinhala-script accuracy. The headroom form regresses the paired loss on Sinhala-script headroom above chance and is reported only to show why we do not rely on it.

Fit	Value
<i>across checkpoints, n = 10</i>	
levels: Romanized on native accuracy	
slope	0.221
95% boot. CI	[+0.113, +0.538]
leave-one-out	0.173 to 0.256
R^2	0.647
p , slope = 0	.0050
p , slope = 1	< .0001
<i>headroom form: loss on headroom</i>	
slope	0.779
95% boot. CI	[+0.457, +0.883]
leave-one-out	0.744 to 0.827
R^2	0.958
p , slope = 0	< .0001
p , slope = 1	.0050
<i>permutation null, Romanized scores shuffled</i>	
levels slope, mean	-0.000
levels R^2 , mean	0.113
levels $\Pr(R^2 \geq \text{observed})$	0.005
headroom slope, mean	+1.000
headroom R^2 , mean	0.936
headroom $\Pr(R^2 \geq \text{observed})$	0.089
<i>across benchmark cells, n = 16</i>	
levels: Romanized on native accuracy	
slope	0.451
95% boot. CI	[+0.359, +0.628]
leave-one-out	0.428 to 0.501
R^2	0.753
p , slope = 0	< .0001
p , slope = 1	< .0001
item-weighted slope	0.476

Romanized side native speakers typed, our text is 0.174 to 0.240 bits per byte harder than theirs, positive for all six checkpoints we could rerun, with a median of 0.219 (Table 16). Our condition is therefore somewhat harder than authentic typing, which makes the downstream gaps we report closer to upper bounds than to lower bounds. This cuts against what the spelling statistics alone would suggest, and we report it because it bounds the claim rather than because it helps it. Two things limit how far that control carries. It was scored on CPU so that it would not compete with the main runs for GPU time, which is why it covers the smaller end of the pool: six of 31 checkpoints, all below 3.2B parameters, and none of the six passes the downstream competence screen. For the checkpoints that do carry the downstream result we instead test the same worry item by item on SOLD, and find

Table 12: Spearman correlations between intrinsic measurements and downstream script effects, over the ten parseable checkpoints, with bootstrap intervals. Levels track the effect, gaps and ratios do not. The intervals are wide because $n = 10$, which is why this is exploratory.

Intrinsic predictor	MMLU gap	SOLD Δ MCC
	ρ 95% CI	ρ 95% CI
Sinhala BPB	-0.83 [-1.00, -0.30]	-0.78 [-1.00, -0.36]
Romanized BPB	-0.85 [-1.00, -0.43]	-0.88 [-1.00, -0.40]
Sinhala BPW	-0.83 [-1.00, -0.31]	-0.78 [-1.00, -0.36]
BPB gap	-0.15 [-0.89, +0.63]	-0.42 [-0.83, +0.33]
PPL ratio	-0.52 [-0.94, +0.28]	-0.41 [-0.96, +0.40]
parameters	+0.58 [-0.06, +0.91]	+0.47 [-0.14, +0.90]

Table 13: Transliteration quality against human references: 4,397 authentic social-media sentence pairs, 450,587 multi-reference words, and a fixed 300,000-sentence sample of a machine-augmented set. Coverage is the share of inputs the method produced any output for.

Method	CER ↓			chrF	Exact	Cov.
	sent.	word	aug.	sent.	word	%
Ours	0.182	0.121	0.112	67.8	33.2	100.00
Web tool $v \rightarrow w$	0.182	0.133	0.114	67.8	30.9	99.67
Aksharamukha	0.191	0.148	0.139	63.7	23.9	100.00
Web tool, raw	0.197	0.176	0.153	63.4	20.6	99.67
uroman	0.228	0.228	0.210	55.7	9.7	100.00

that how closely an item’s Romanized spellings match attested human spellings does not predict its Romanized penalty, with a Spearman correlation of 0.02 ($p = 0.31$, $n = 2,447$) and terciles losing 5.0, 7.2 and 7.3 points from least to most attested (Table 17). The intrinsic results do not share this limitation at all, since their Romanized side was typed by native speakers, and they show the same flattening. A human-written Romanized version of even a few hundred SinhalaMMLU items would still settle the question better than any of this.

I Full limitations

One language. Everything here is Sinhala. The mechanism we describe, a floor effect that makes robustness gaps proportional to formal-script ability, should generalize to any language with a dominant Romanized register, but we have not shown that. Hindi, Nepali, Telugu, Urdu, Tamil and Arabic dialects are the obvious tests, and Khullar et al. [4] report compatible magnitudes for Hindi, Nepali, Telugu, Kannada, Marathi and Punjabi.

The Romanized side of the downstream sets is synthetic. See §H. We transliterate rather than collect human-typed Romanized versions, because no such parallel resource exists at this scale, and we measured that our text is the harder end of the range rather than assuming it is neutral.

Checkpoint size is confounded with everything else. Our 31 checkpoints differ in family, tokenizer, training corpus and instruction-tuning status at the same time. The association between parameter count and Sinhala-script bits per byte is not a scaling law, and our own fertility analysis is direct evidence that family-level confounds are present.

We cannot exclude contamination. The training corpora of these checkpoints are not fully auditable, so we cannot rule out that public evaluation sets, or Sinhala-script material closely related to them, appeared during pretraining. If they did, the Sinhala-script side would be inflated and the gaps we report overstated. This applies to any evaluation of pretrained models on public benchmarks and we have no way to bound it here.

The intrinsic-to-downstream link is exploratory. Ten checkpoints, no held-out validation, no pre-registration, and bootstrap intervals wide enough to reach -1.00 (§G).

Table 14: Share of word tokens whose Romanized spelling is one a human supplied for that Sinhala word in the Swa-bhasha lexicon. “tokens” counts the word tokens the lexicon covers and “cov.” the share of all Sinhala word tokens that is. “norm.” collapses runs of the same Latin vowel on both sides. The first row is the reference point: it applies the same measure to text native speakers typed.

Text	tokens	cov. %	attested %	
			raw	norm.
parallel, typed by people	2,797	96.1	83.4	84.2
parallel, ours	2,905	96.0	64.3	88.6
SOLD, ours	38,544	90.3	59.5	87.0
Global PIQA, ours	2,244	96.3	63.9	86.9

Table 15: The twelve most frequent word-level disagreements with the human-typed side of the parallel corpus that survives normalizing the long-vowel convention. Most are vowel reduction or abbreviation on the human side.

Ours	Human	n
<i>mee</i>	<i>meh</i>	33
<i>man</i>	<i>mn</i>	30
<i>daen</i>	<i>dan</i>	28
<i>naae</i>	<i>na</i>	26
<i>mata</i>	<i>matta</i>	19
<i>ee</i>	<i>a</i>	15
<i>naae</i>	<i>nah</i>	12
<i>ban</i>	<i>bn</i>	12
<i>nam</i>	<i>nm</i>	11
<i>gaena</i>	<i>gana</i>	10
<i>thamayi</i>	<i>thamai</i>	10
<i>waedak</i>	<i>wadak</i>	9

The mixed-script condition is weaker evidence than the rest. That corpus is authentic mixed-script text rather than a transliteration of the parallel set, so it is not parallel to either script condition. It supports a descriptive statement about how mildly script mixing perturbs cost, and nothing paired.

Open-weight SLMs only. We restrict downstream evaluation to open-weight small language models (SLMs) up to 14.7B parameters, the class of models feasible for local hosting and on-device deployment in low-resource communities. Frontier proprietary models are much stronger on Sinhala [10] and therefore have far more headroom, so the flattening slope predicts larger absolute losses for them. That prediction is untested here.

Zero-shot only. One template, zero-shot, greedy. This measures what a user gets out of the box, not the ceiling. Few-shot prompting, transliterate-then-infer, or Romanized fine-tuning could all narrow the gap, and we do not measure by how much.

Two checkpoints cannot do the task. LaMini-GPT-1.5B and TinyLlama-1.1B-Chat produce unparseable output on most Sinhala-script items of at least one task. Their apparent improvement under romanization is an artifact of that failure, not a script effect, which is why the competence screen exists.

J Ethics

Content warning. §4 and Appendix E discuss an offensive-language dataset. We quote no offensive text anywhere in this paper and report only aggregate metrics over it.

Data. All datasets are public research releases used for evaluation only. Global PIQA forbids training on it or on synthetic data seeded from it, and we do neither. SinhalaMMLU is released under a license that forbids distributing adaptations, and its authors asked that the full set not be made public, so our Romanized version of it stays local. SOLD contains offensive social media

Table 16: Our transliteration against human typing on the same 500 sentences. Δ is ours minus human bits per byte, positive meaning our text is harder. “PPL ratio” is our perplexity over the human perplexity, below one for every checkpoint even though bits per byte is higher, because our spellings occupy more tokens. The Sinhala-script and human columns reproduce the recorded run to within 0.11%. ‡ marks checkpoints also evaluated downstream. Six of 31 checkpoints, all below 3.2B and none passing the competence screen, so this bounds the direction rather than settling it.

Checkpoint	bits per byte				PPL	
	B	Sinh.	human	ours	Δ	ratio
OPT-350M	0.35	2.238	3.986	4.195	+0.208	0.85
Pythia-410M	0.41	1.350	3.707	3.937	+0.230	0.80
BLOOM-560M	0.56	2.287	3.810	4.029	+0.220	0.62
TinyLlama-1.1B-Chat‡	1.10	1.468	3.922	4.096	+0.174	0.73
Qwen1.5-1.8B	1.80	1.601	3.573	3.791	+0.218	0.73
SmolLM3-3B‡	3.10	1.878	4.147	4.387	+0.240	0.75
median	—	—	—	—	+0.219	0.74

Table 17: SOLD items split into terciles by how often their Romanized spellings are attested in the human lexicon, pooled over the five competent checkpoints. If our transliterator’s idiosyncrasies drove the Romanized penalty, the low-attestation tercile would lose most. It loses least. The tercile with the smallest gap also has the lowest Sinhala-script accuracy, which the flattening of §4 already predicts.

Attest.	items	att. %	S %	R %	Δ	95% CI
low	822	40.8	61.0	56.0	5.0	[3.4, 6.6]
middle	925	60.7	61.9	54.7	7.2	[5.6, 8.8]
high	700	78.9	60.4	53.2	7.3	[5.4, 9.1]

text, already pseudonymised upstream with user mentions replaced by placeholders, and we neither collected nor re-annotated any of it. We add no new human annotation and no new data collection, so no ethics approval was required.

Models. All 31 checkpoints were used for evaluation only, with no fine-tuning, redistribution or derivative model release, which is within the terms of each family’s license, including the Llama Community License, the Gemma Terms of Use and the BigScience RAIL license.

Risk. The practical risk our results speak to is deployment. A moderation or support system validated on Sinhala-script text and serving Romanized users will look far better in evaluation than in the field, and on SOLD that failure is invisible in accuracy while more than half the usable signal is gone. Publishing the size of that gap, and a screen that stops averaging from hiding it, is the useful response. Our transliterator collapses several Sinhala consonant distinctions, matching casual Singlish practice, so it is a tool for building evaluation data and not for rendering names or other text where those distinctions carry meaning.

K Licenses and released artifacts

What we release. The repository at <https://github.com/sehan-dissanayake/sinhala-script-robustness> contains the transliterator, the evaluation harness, all analysis code, the frozen result files and the scripts that regenerate every number, table and figure in this paper. The transliterator is also installable from PyPI.

Datasets. SinhalaMMLU [10] is CC BY-NC-ND 4.0 and gated. We received the full set from its authors, who ask that it not be redistributed, and the no-derivatives clause forbids distributing our Romanized adaptation, so we release the transformation code and the item identifiers instead of the items. SOLD [2] carries no explicit license file and we use its published test split for evaluation only. The Sinhala split of Global PIQA [11, 12] is CC BY-SA 4.0 and forbids training, and we use it for evaluation only. The parallel and mixed-script corpora come from Rajapakse and Weerasinghe [5], and the human-romanized reference data from the Swa-bhasha Resource Hub [13].

Tools. Aksharamukha [30] v2.3 is AGPL-3.0. As its license requires, we state that this project uses the universal romanizer software ‘uroman’ written by Ulf Hermjakob, USC Information Sciences Institute (2015-2020) [31].